SIDANG TUGAS AKHIR

KLASIFIKASI OSTEOARTRITIS LUTUT DENGAN *TEMPORAL CONVOLUTIONAL NETWORK* BERBASIS *MARKERLESS VISION* PADA VIDEO *GAIT*

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DEPARTEMEN MATEMATIKA
FAKULTAS SAINS DAN ANALITIKA DATA
INSTITUT TEKNOLOGI SEPULUH NOPEMBER

528 juta

Penduduk di dunia yang menderita Osteoarthritis (Langworthy et al., 2024)



Sumber: halodoc.com



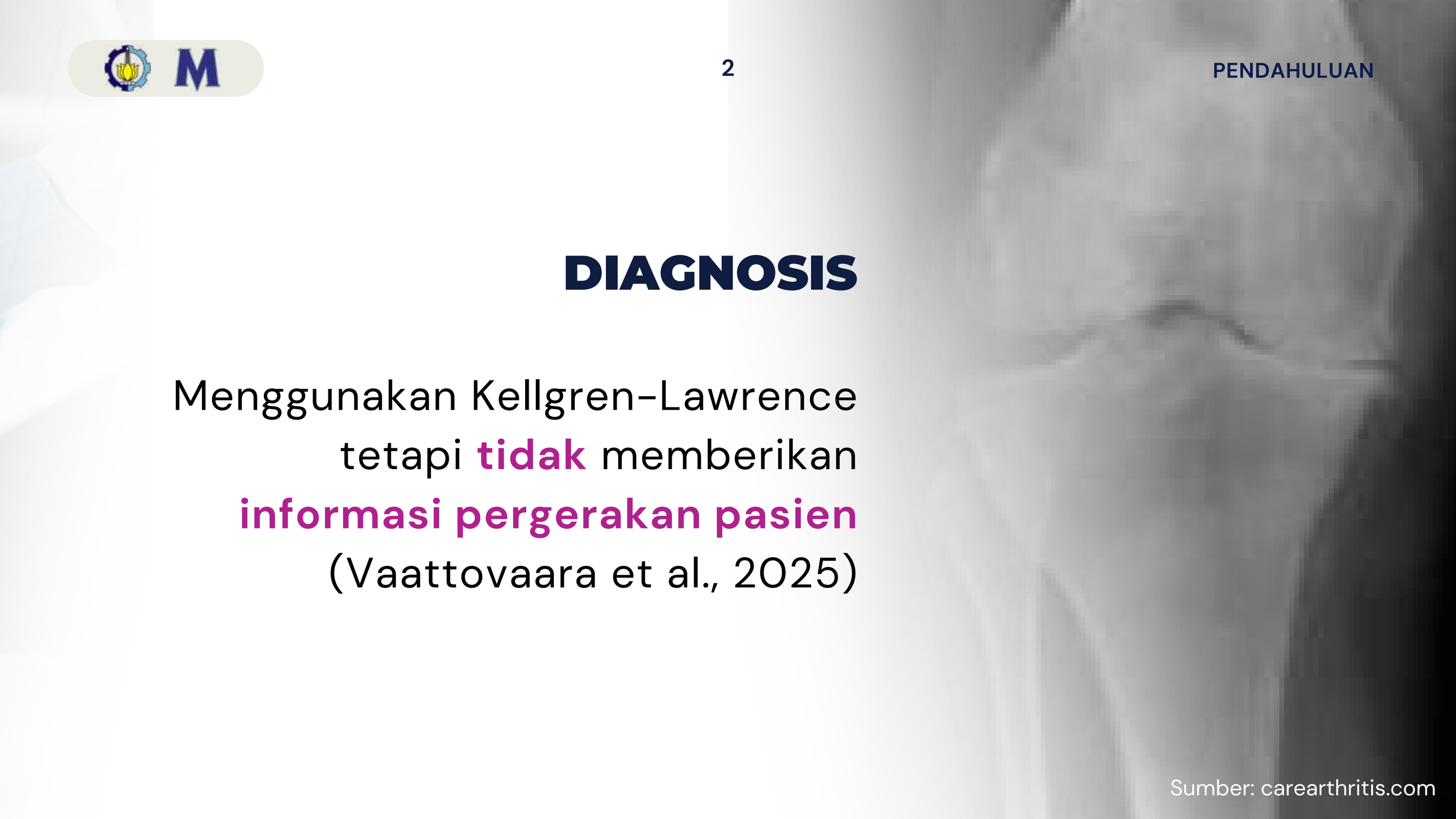
Sumber: ekahospital.com

STRATEGI PENGOBATAN

Bergantung pada **derajat keparahan** (Geng et al., 2023)

DIAGNOSIS

Menggunakan Kellgren-Lawrence tetapi **tidak** memberikan **informasi pergerakan pasien** (Vaattovaara et al., 2025)

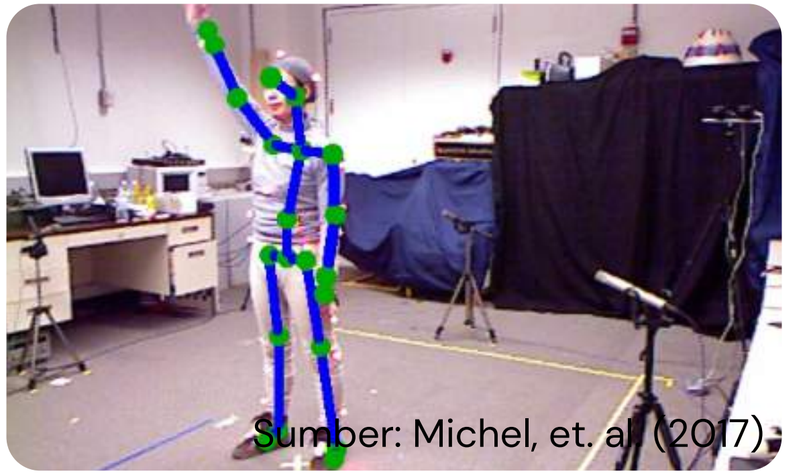


Sumber: carearthritis.com



ANALISIS *GAIT*

Analisis Gaya Berjalan



Sumber: Michel, et. al. (2017)

Markerless vision menjadi **alternatif** dari *marker-based* dan sensor (Wang et al., 2024)

Klasifikasi osteoartritis lutut **tanpa** memanfaatkan **informasi temporal** (Ben Hassine et al., 2024)

Temporal Convolutional Network (TCN) (Lea et al., 2017).

BATASAN MASALAH

DATASET



“Vision-Based Gait Dataset for Knee Osteoarthritis and Parkinson’s Disease Analysis with Severity Levels” (Kour et al., 2020)

KELAS KLASIFIKASI

- Early (EL)
- Moderate (MD)
- Severe (SV)

MARKERLESS VISION

Human Pose Estimation

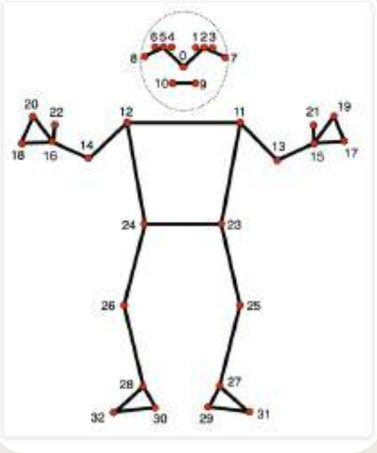
Tugas visi komputer dalam identifikasi anatomi manusia dalam koordinat (Szeliski, 2022).

Ekstraksi Fitur

Ekstraksi fitur kinematika dari *keypoints* berupa sudut sendi lutut dengan persamaan:

$$angle[deg] = \arctan\left(\frac{y_3 - y_2}{x_3 - x_2}\right) - \arctan\left(\frac{y_1 - y_2}{x_1 - x_2}\right)$$

(Hassine et. al., 2024)

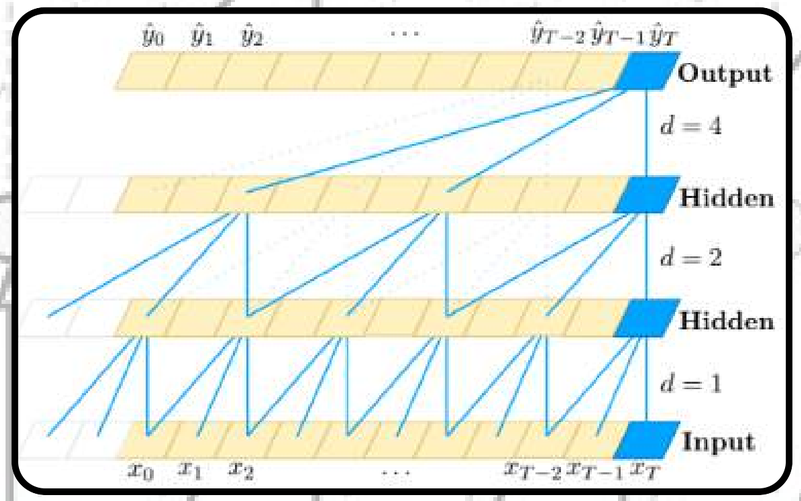


MediaPipe

33 *keypoints* MediaPipe (Google AI Edge, 2025)

TEMPORAL CONVOLUTIONAL NETWORK (TCN)

deep learning **berbasis 1D *fully convolutional network* (1D-FCN)** untuk **tugas temporal**



Sumber: Bai, et. al. (2018)

TEMPORAL CONVOLUTIONAL NETWORK (TCN)

Causal Convolution

Output pada waktu T hanya terpengaruh oleh elemen dari layer sebelumnya

$$y_t = f(x_0, \ldots, x_t)$$

Dilated Convolution

Memperluas *receptive field (RF)* secara eksponensial dengan faktor dilatasi d_l

$$F(s) = \sum_{i=0}^{k-1} \mathbf{w}(i) \cdot x_{s-d \cdot i}.$$

Residual Block

Memungkinkan input asli (x) ditambahkan ke output dari lapisan konvolusi (F(x)) sebelum aktivasi

$$o = \text{Activation}(x + \mathcal{F}(x))$$

PRAPEMROSESAN DATA

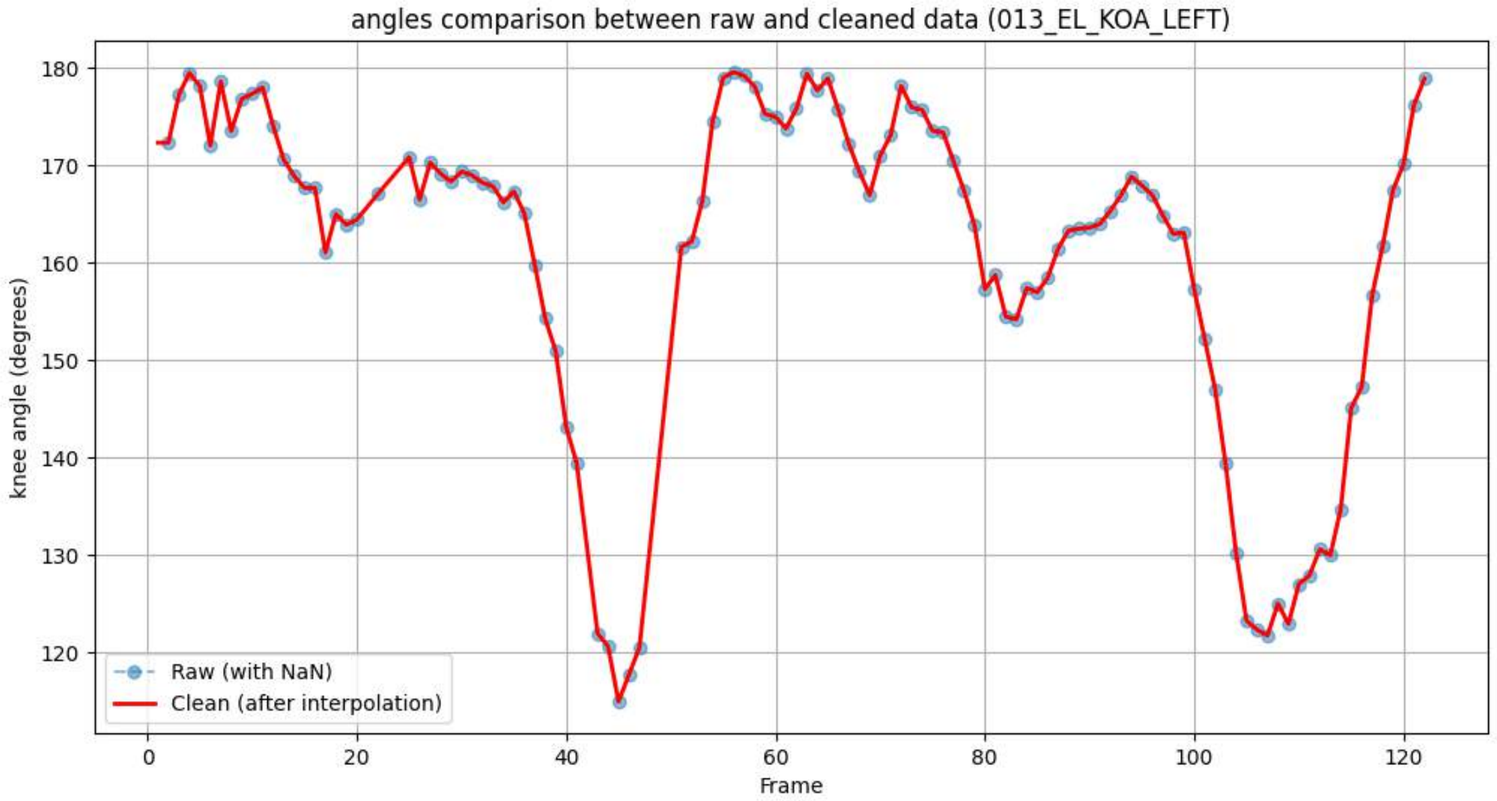
Estimasi Pose 2D



PRAPEMROSESAN DATA

Ekstraksi Fitur dan Pembersihan Data

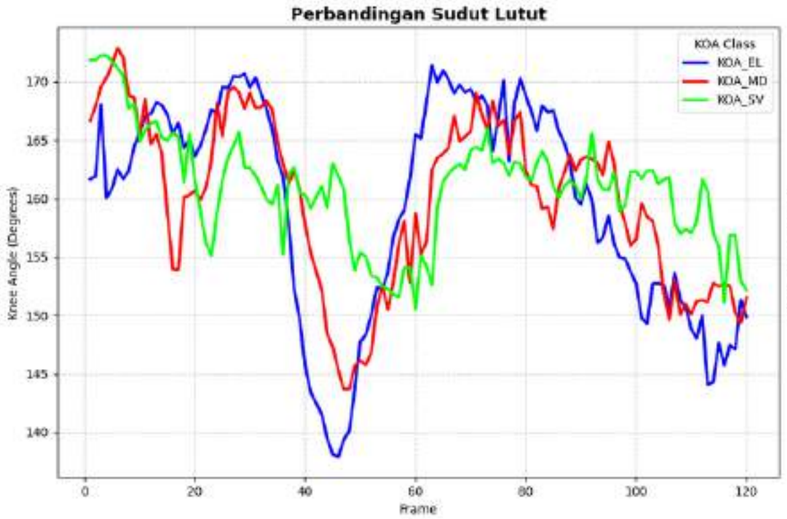
```
frame,knee_angle,cleaned_angles
1,,172.32575980397854
2,172.32575980397854,172.32575980397854
3,177.24590707586995,177.24590707586995
4,179.47962724485717,179.47962724485717
5,178.13260817698443,178.13260817698443
6,172.03664383342596,172.03664383342596
7,178.59645219338367,178.59645219338367
8,173.52146701935186,173.52146701935186
9,176.8182863565415,176.8182863565415
10,177.3416920625219,177.3416920625219
11,177.98258850266362,177.98258850266362
12,174.0711249179031,174.0711249179031
13,170.60833682431527,170.60833682431527
14,168.9174551411952,168.9174551411952
15,167.68701869503963,167.68701869503963
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18,165.00259004021217,165.00259004021217
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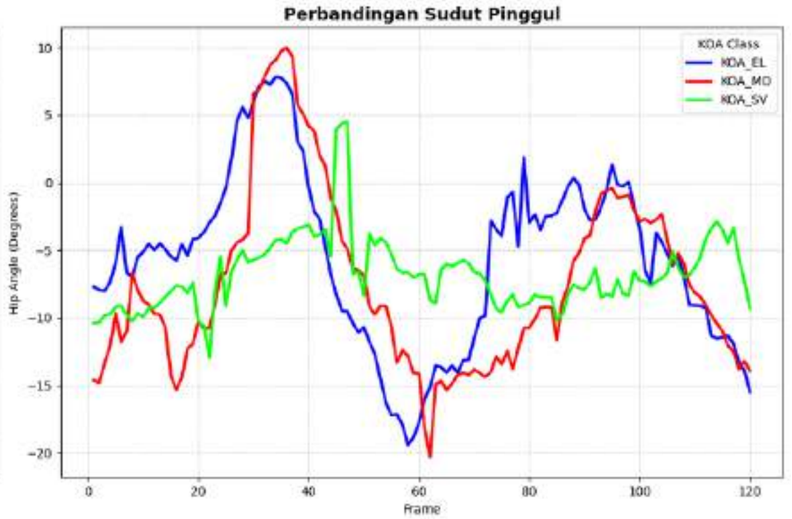


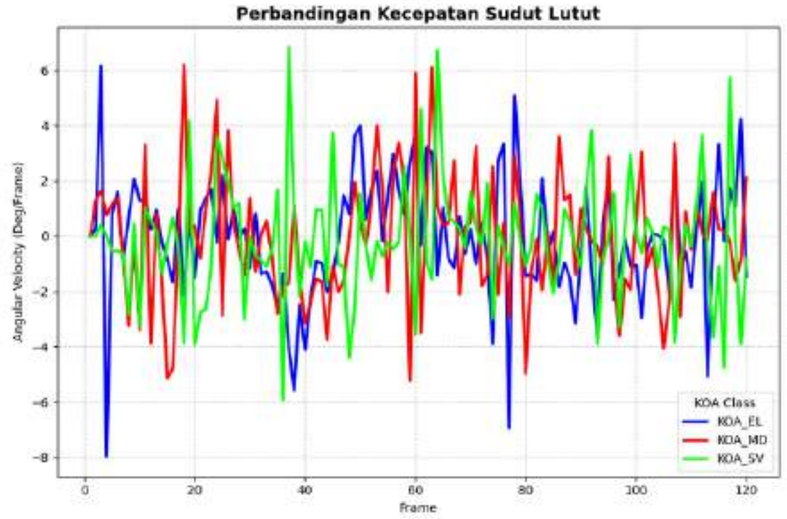
perbandingan data mentah dengan data setelah diinterpolasi

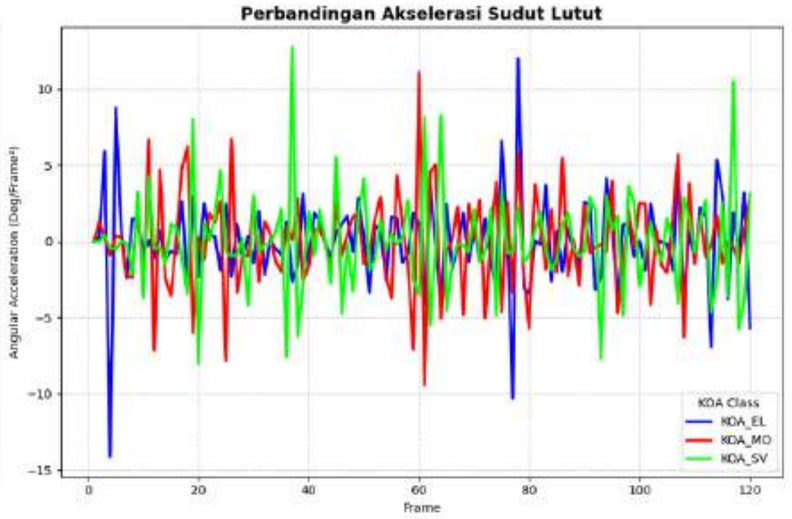
PRAPEMROSESAN DATA

Ekstraksi Fitur dan Pembersihan Data





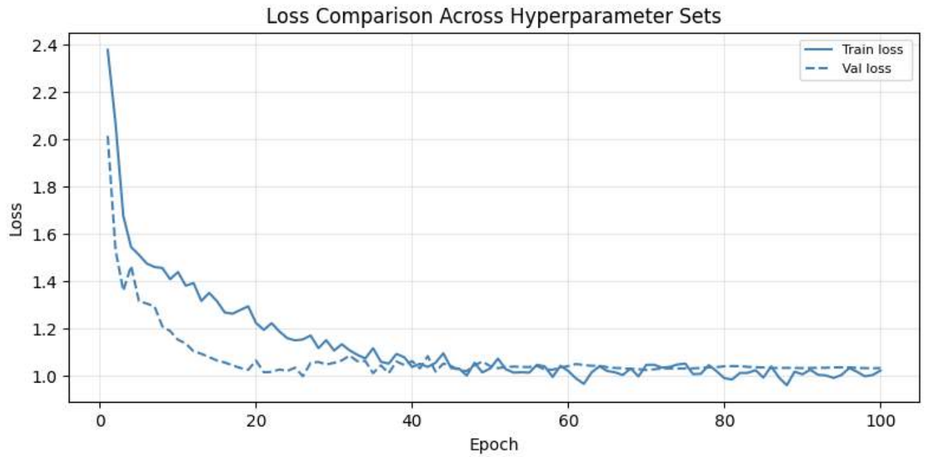


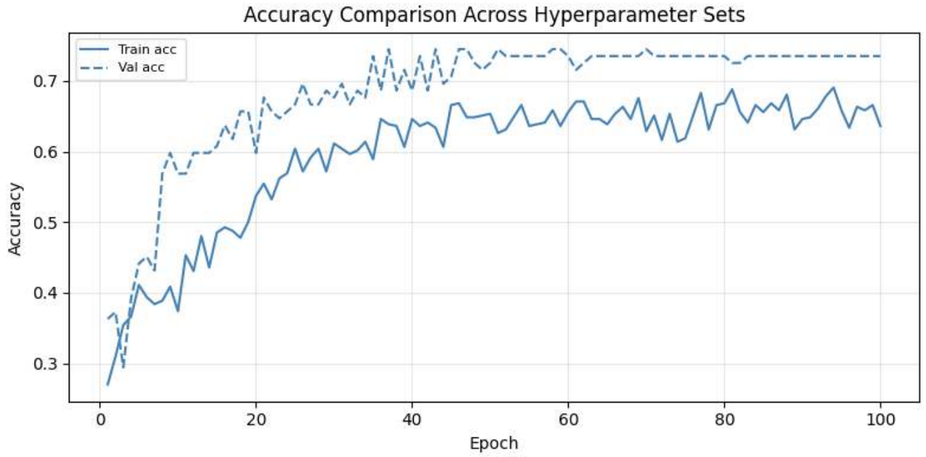


- sudut lutut
- sudut panggul
- kecepatan sudut lutut
- akselerasi sudut lutut

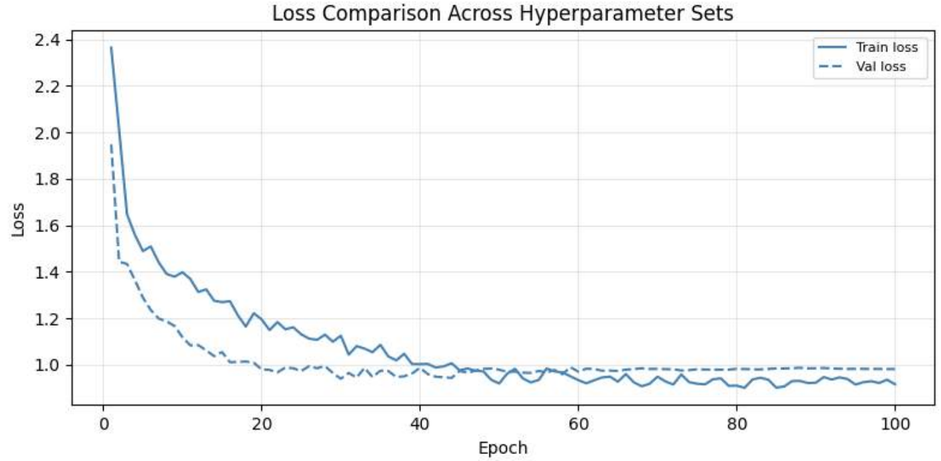
SKENARIO 3 - FUNGSI LOSS & *OVERSAMPLING*

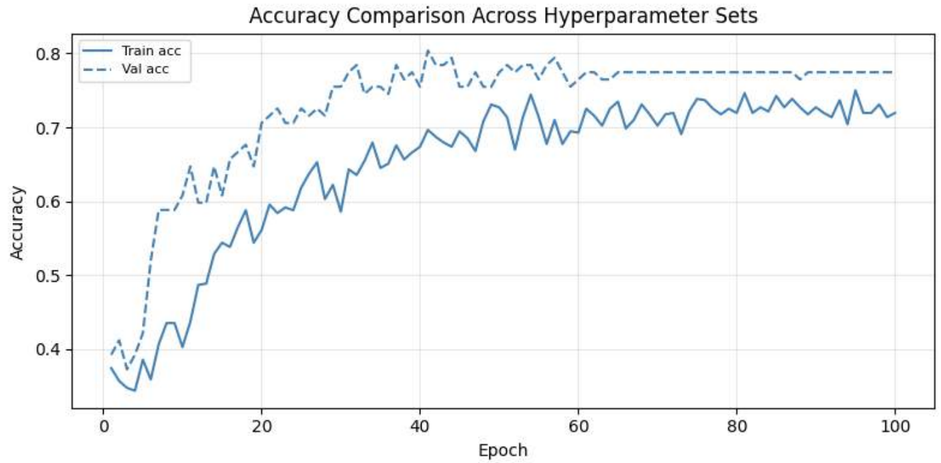
Sparse Categorical Cross Entropy + Tanpa Oversampling





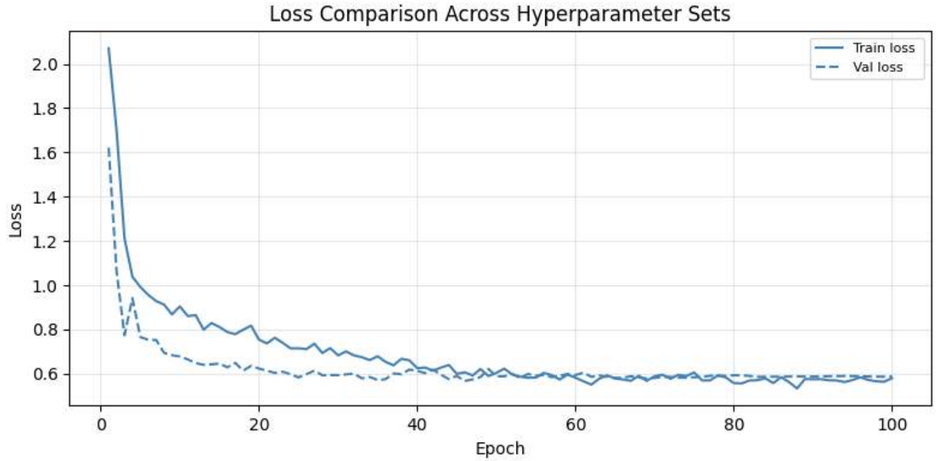
Sparse Categorical Cross Entropy + Oversampling

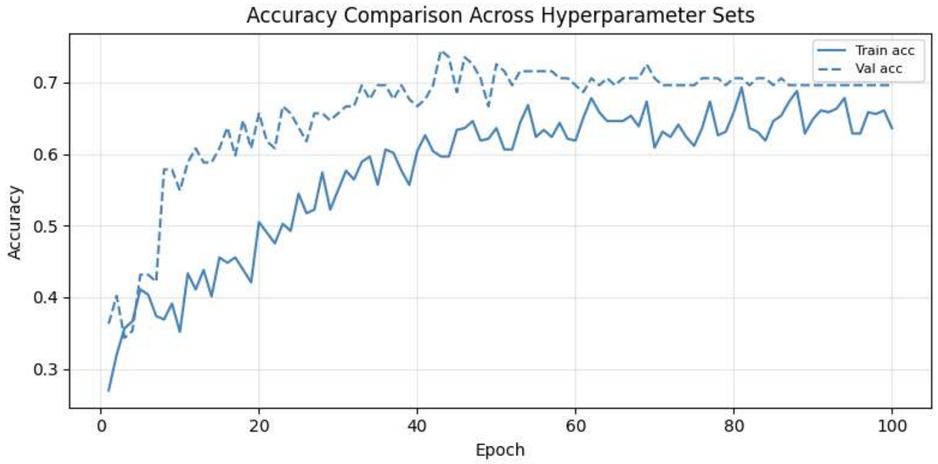




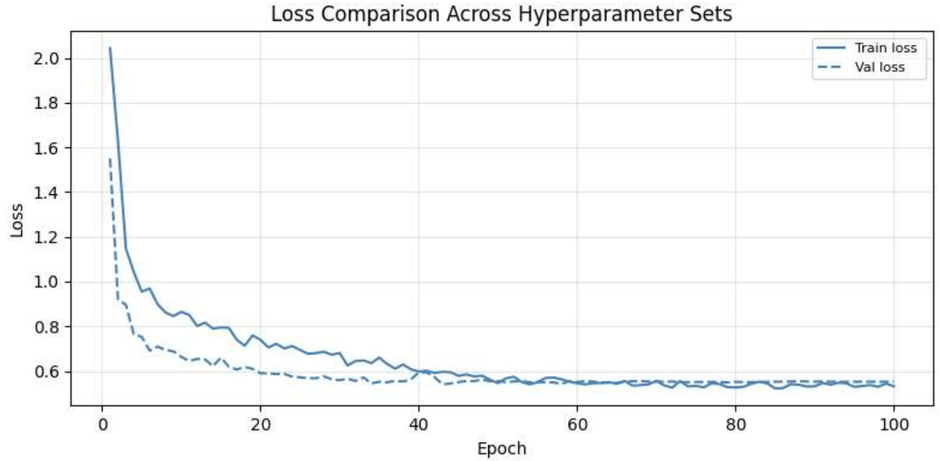
SKENARIO 3 - FUNGSI LOSS & *OVERSAMPLING*

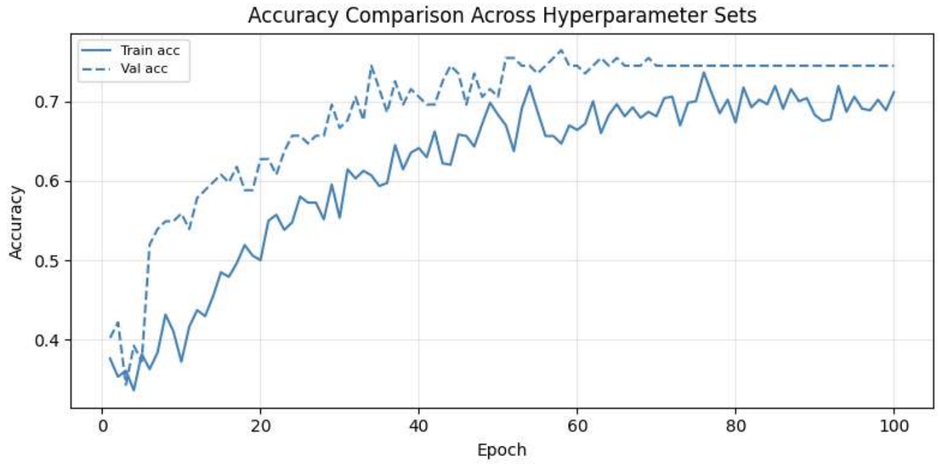
Sparse Categorical Focal Loss + Tanpa Oversampling





Sparse Categorical Focal Loss + Oversampling





SKENARIO 3 - FUNGSI LOSS & *OVERSAMPLING*

Analisis Hasil

Skema	Train Acc	Val Acc
Skema 1	70.54%	74.51%
Skema 2	72.90%	80.39%
Skema 3	70.05%	74.51%
Skema 4	**74.62%**	**76.47%**

Skema 4 menunjukkan **stabilitas generalisasi terbaik** dengan **selisih performa yang paling minimal,** yakni hanya sebesar **–1,85%.** Gap yang kecil pada Skema 4 ini menandakan bahwa model memiliki **keseimbangan optimal** antara kemampuan mempelajari pola data latih (fitting) dan kemampuan memprediksi data baru.

Skema 4 adalah model terbaik

HASIL MODEL TERBAIK

Precision, Recall, F1 Score, Accuracy

Kelas	Precision	Recall	F1-Score	Support
KOA_EL	0.50	0.19	0.28	26
KOA_MD	0.52	0.81	0.63	31
KOA_SV	0.98	0.96	0.97	53
Macro avg	0.67	0.65	0.63	110
Weighted avg	0.74	0.74	0.71	110
Akurasi keseluruhan				**0.7364**

Akurasi **73,64%**

Precision 67%

Recall 65%

F1-Score 63%

Confusion Matrix

